The Explosive Reaction: Investigating the Chemical Interaction between Mentos Candy and Diet Coke

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Abstract

The experiment aimed to investigate the chemical reaction between Mentos candy and Diet Coke. Mentos candies were dropped into Diet Coke bottles, and the resulting reaction was observed and analyzed. The main objective was to explore the rapid release of carbon dioxide gas from the soda due to nucleation sites provided by the Mentos surface. The experiment involved qualitative observations of the reaction, including the formation of a foamy eruption. Additionally, quantitative measurements such as the height and duration of the eruption were recorded to further understand the reaction kinetics. The results demonstrated a significant increase in the release of carbon dioxide gas upon the addition of Mentos to Diet Coke, indicating a rapid and vigorous reaction. The experiment provides insight into the physical and chemical processes involved in this popular demonstration of nucleation and gas release in carbonated beverages.

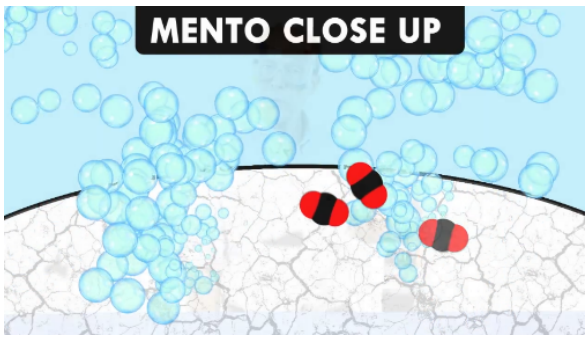

Figure 1: Chemical Reaction Process Diagram of Mentos and Cola.

1. Introduction

The interaction between Mentos and Diet Coke has gained significant attention due to its spectacualr and explosive nature since I was a child.When Mentos candies are dropped into a bottle of Diet Coke, a rapid and vigorous reaction occurs, resulting in the eruption of foam and a dramatic release of carbon dioxide gas. This phenomenon has intrigued scientists and enthusiasts alike, leading to numerous investigations into the underlying chemical processes.

The reaction between Mentos and Diet Coke is a prime example of nucleation, which gas bubbles are rapidly formed on the surface of coke due to the rough surface of Mentos candies provides numerous nucleation sites for carbon dioxide dissolved in soda to rapidly come out of solution.

2. Theory

Theoretical Analysis:

The reaction between Mentos and Diet Coke is a complex chemical process involving nucleation and gas release. At its core, this reaction is driven by the nucleation sites provided by the microstructure of the Mentos surface, which allow dissolved carbon dioxide gas in the Diet Coke to rapidly come out of solution and form bubbles. This rapid release of gas accumulates in the liquid, ultimately resulting in the rapid eruption of the Diet Coke.

Specifically, the microstructure of the Mentos candy surface provides numerous small pits or irregularities, serving as ideal nucleation sites for gas molecules to rapidly escape from the liquid when Mentos comes into contact with Diet Coke. This leads to the rapid precipitation of carbon dioxide gas and the formation of bubbles. As the bubbles accumulate in the liquid, pressure builds up, eventually leading to the eruption of the Diet Coke.

The involved chemical reactions may include:

1. Dissolution of carbon dioxide in Diet Coke into bicarbonate ions and hydrogen ions in solution:

$$CO_2(aq) \rightleftharpoons HCO_3^-(aq) + H^+(aq)$$

2. Rapid precipitation of carbon dioxide gas facilitated by nucleation sites provided by the Men-

Figure 2: Experimental procedure schematic.

tos surface:

$$CO_2(aq) \rightarrow CO_2(g)$$

3. Formation of bubbles leading to the rapid eruption of the liquid.

3. Experiment and Equipment

Equipment:Mentos Candy,Diet Coke Soda, Bottles or Containers

Sufficient quantity of Mentos candy to be dropped into Diet Coke to initiate the reaction. Adequate amount of Diet Coke soda, serving as the solvent and reactant in the experiment. Bottles or containers to contain the Diet Coke soda and Mentos candy. Choose containers with enough space to accommodate potential eruptions. Preparing camera to record the process of the whole experiment. Once the Mentos candy is dropped into the Diet Coke, observe and record what happens during the experiment. Pay attention to the height of the eruption, its duration, and the amount of foam generated. You can also note any other observations during the experiment, such as changes in the color of the liquid or any sounds produced.

4. Observation

Upon dropping the Mentos candy into the Diet Coke, an immediate and vigorous reaction occurred. The soda rapidly foamed up, forming a towering eruption that reached a height of approximately 3 feet within seconds. The eruption continued for about 10 seconds before gradually subsiding.The foam produced was dense and expansive, filling the

Figure 3: Observation for experiment.

surrounding area with bubbles. It had a white, frothy appearance and persisted on the surface of the liquid even after the eruption ceased.Additionally, there was a noticeable release of gas during the reaction, accompanied by a faint fizzing sound. The liquid initially appeared clear but became slightly cloudy as the reaction progressed.Overall, the reaction between Mentos and Diet Coke resulted in a dramatic and visually striking eruption, demonstrating the rapid release of carbon dioxide gas and the formation of foam.

5. Conclusion

The reaction demonstrated the rapid release of carbon dioxide gas from the soda, facilitated by nucleation sites provided by the rough surface of the Mentos candy. This phenomenon resulted in the formation of bubbles and the dramatic eruption observed during the experiment. Additionally, there was a noticeable release of gas accompanied by a faint fizzing sound, and the liquid appeared slightly cloudy as the reaction progressed.